

Heatmap visualization showing the enrichment of 35 biological processes across five clusters (clust1 to clust5). The color scale ranges from 0 (white) to 8 (dark red). Asterisks (\*) indicate significant enrichment.

**Biological Processes (Rows):**

- Type I diabetes mellitus
- Graft-versus-host disease
- Allograft rejection
- Autoimmune thyroid disease
- Diabetic cardiomyopathy
- Oxidative phosphorylation
- Mitophagy – animal
- Insulin signaling pathway
- Ubiquitin mediated proteolysis
- Nucleocytoplasmic transport
- mRNA surveillance pathway
- Autophagy – animal
- Protein processing in endoplasmic reticulum
- Spliceosome
- Cell cycle
- ATP-dependent chromatin remodeling
- Coronavirus disease – COVID-19
- Ribosome
- Antigen processing and presentation
- Phagosome
- Asthma
- Viral myocarditis
- Complement and coagulation cascades
- Chemical carcinogenesis – DNA adducts
- Thermogenesis
- Chemical carcinogenesis – reactive oxygen species
- Non-alcoholic fatty liver disease
- Pathways in cancer
- Cytokine–cytokine receptor interaction
- Retrograde endocannabinoid signaling
- Cytoskeleton in muscle cells
- Parkinson disease

**Clusters (Columns):** clust5, clust3, clust1, clust2, clust4

**matrix\_2 Legend:**

- 8 (Dark Red)
- 6 (Red)
- 4 (Light Red)
- 2 (White)
- 0 (White)

C

0

4

2

1

0